

11 Nutrient interaction and root zone map

Interactions are best used as a checklist of what to measure together. They do not create universal ratios or treatment thresholds.

Nutrient	Read together with	Main question
Nitrogen	Water oxygen organic matter salinity	Is supply synchronized with demand or being tied up leached or lost in saturated soil?
Phosphorus	pH zinc iron boron roots	Is P unavailable or excessive and is it contributing to micronutrient imbalance?
Calcium	Potassium magnesium sodium water flow	Is the issue supply cation competition soil structure or poor delivery through roots?
Sulfur	Potassium pH EC water sulfate	Is sulfate coming from water or amendments and does the salt budget explain the pattern?
Manganese	pH redox iron zinc magnesium calcium	Is Mn reporting a wet acidic or otherwise stressed root zone?
Iron	pH bicarbonate drainage roots	Is iron present but unavailable because of chemistry or root failure?
Zinc	Phosphorus pH bicarbonate boron	Is uptake being limited by P buildup alkaline chemistry or root conditions?
Magnesium	Potassium calcium sodium CEC	Is Mg sufficient within the measured cation and water setting without chasing a fixed ratio?
Boron	Water salinity chloride sodium P Zn	Is the symptom deficiency toxicity or combined salt injury and what is the continuing source?

How Phase 1 should guide Phase 2

- **Use the model to prioritize inspection.** Extra Trees outperformed additive splines, showing that nutrient background and nonlinear context matter.
- **Keep states continuous.** The high Ca Mg Mn Fe and low K P Zn B pattern—and its reverse—describe recurring backgrounds, not permanent grove types.
- **Keep interaction claims in their lane.** K by S earned research follow-up; K by Ca and Ca by Mg are cation context; P by B and Zn by B are candidates; Mn is a diagnostic context signal.
- **Do not infer causality from prediction.** A model can improve a forecast while still reflecting crop load, irrigation, soil, root condition, season, or management history.

Decision rule When a model flag appears, ask what additional soil, water, root, and management evidence would distinguish the competing explanations. Treat only after the process—not merely the association—is reasonably supported.